PRACTICAL 14-A

Object Oriented Programming

AIM:

To assemble and test a small layered Java application by creating a model, a service that manages data using an ArrayList, and a menu-driven main class; then write JUnit tests to verify successful additions and invalid-action handling.

CODE SCREENSHOT:

LayeredApp.java 1

14-A > LayeredApp.java > { } app

```java
package app;

import service.TaskService;
import java.util.Scanner;

public class LayeredApp {
    public static void main(String[] args) {
        TaskService service = new TaskService();
        try (Scanner sc = new Scanner(System.in)) {
            while (true) {
                System.out.println("\n1. Add task  2. List tasks  0. Exit");
                System.out.print("Choice: ");
                String choice = sc.nextLine().trim();
                try {
                    switch (choice) {
                        case "1" -> {
                            System.out.print("Task title: ");
                            service.performAction("ADD", sc.nextLine());
                            System.out.println("Task added.");
                        }
                        case "2" -> service.performAction("LIST", "");
                        case "0" -> {
                            System.out.println("Goodbye.");
                            return;
                        }
                        default -> service.performAction(choice, "");
                    }
                } catch (IllegalArgumentException ex) {
                    System.out.println(ex.getMessage());
                }
            }
        }
    }
}

```

Code Screenshot 1

Task.java 1 ×

14-A > Task.java > Task

```java
package model;

public record Task(int id, String title) {
    public Task {
        if (title == null || title.isBlank()) {
            throw new IllegalArgumentException("Task title cannot be blank");
        }
    }
}

```

Code Screenshot 2

TaskService.java 1 ×

14-A > TaskService.java > ...

```java
package service;

import model.Task;
import java.util.ArrayList;
import java.util.List;

public class TaskService {
    private final List<Task> tasks = new ArrayList<>();
    private int nextId = 1;

    public Task addTask(String title) {
        Task task = new Task(nextId++, title);
        tasks.add(task);
        return task;
    }

    public int count() {
        return tasks.size();
    }

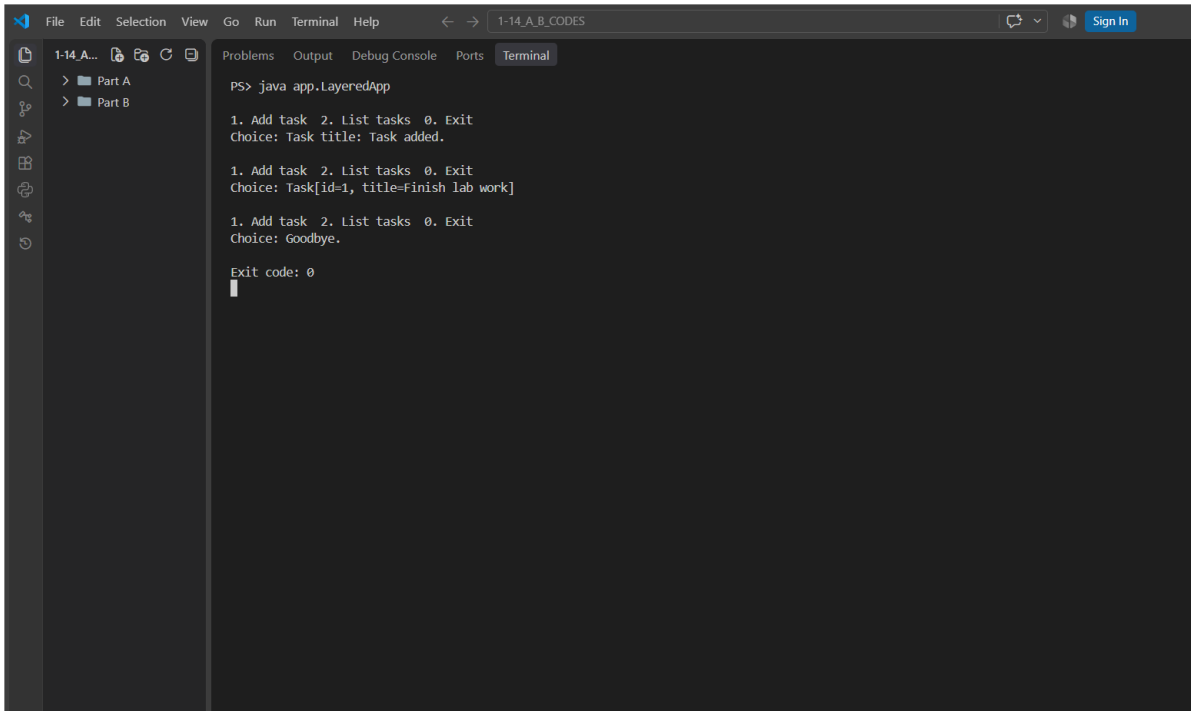
    public List<Task> getAll() {
        return List.copyOf(tasks);
    }

    public void performAction(String action, String value) {
        if ("ADD".equalsIgnoreCase(action)) {
            addTask(value);
        } else if ("LIST".equalsIgnoreCase(action)) {
            getAll().forEach(System.out::println);
        } else {
            throw new IllegalArgumentException("Invalid action: " + action);
        }
    }
}

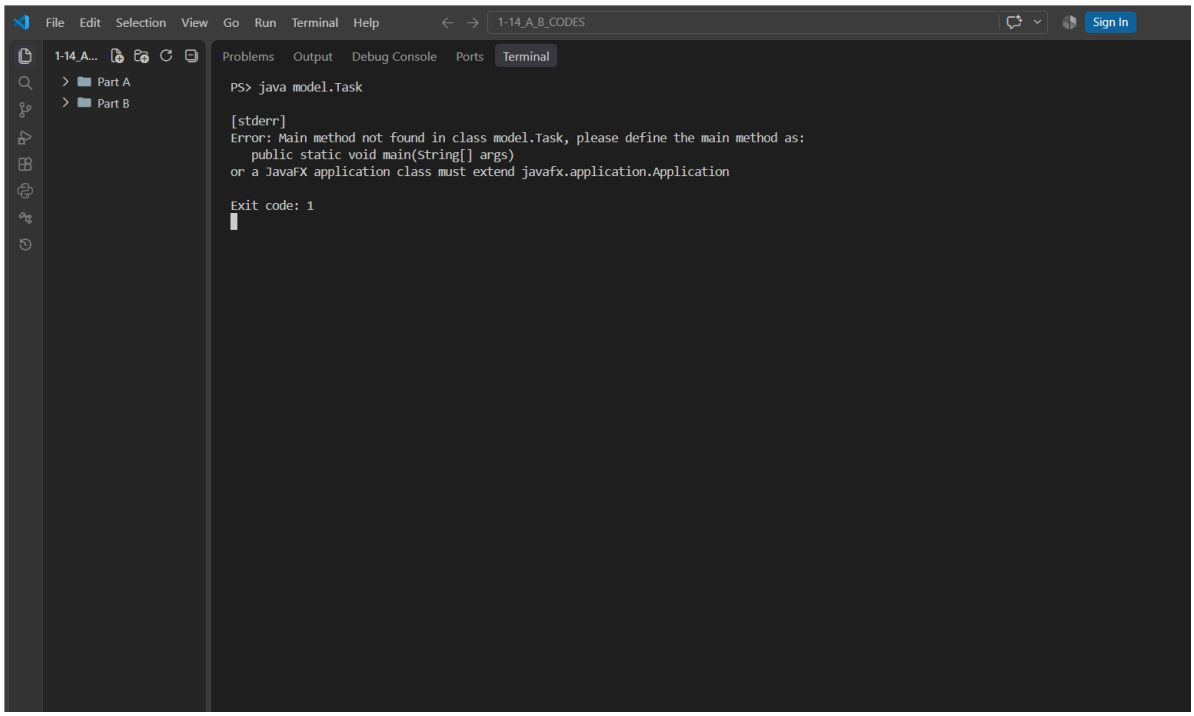
```

Code Screenshot 3

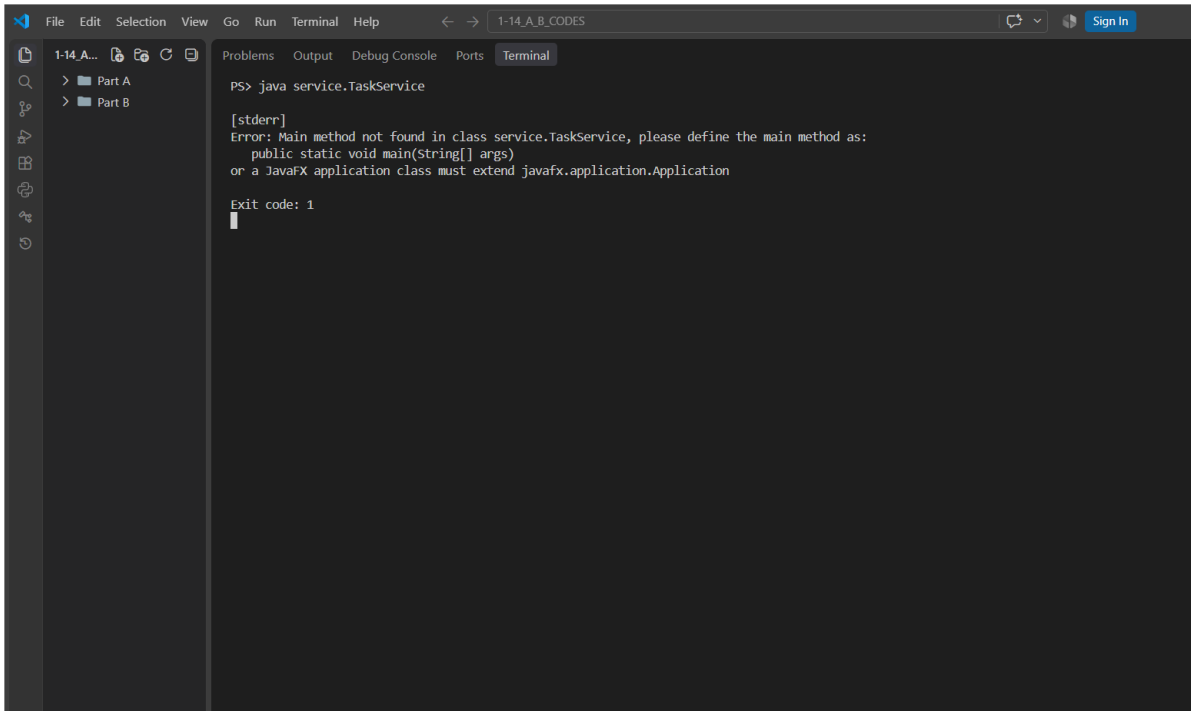
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

The student can assemble a small layered Java application, write basic JUnit tests for expected behaviour and exceptions, and understand how packaging and persistence tests support reliable application delivery.